

Machine Learning: History and Tools

Machine Learning Applications in Life Sciences and Biotechnology

1. How is machine learning used in this environment, and what does it do that humans can't (or would be too time intensive)?

When reading the 8 case studies of machine learning in life sciences and biotechnology there were various ways machine learning comes to our aid. Of the 8 different case studies, we'll focus on a few.

The first case study focused on the development of microbiome therapy. The human microbiome has a massive impact on human health. Imbalances can promote inflammatory bowel disease (IBD) and disorders like Parkinson's disease. Machine learning is used to help predict microbiome imbalances that cause motor neuron diseases like Parkinson's. Using supervised ML and reinforced learning, these algorithms can detect deficiencies and learn how to personalize therapies based on drug knowledge such as dosage, solubility, stability and administration.

Another interesting case was mental illness predictions. With about 10% of the global population suffering from mental illness, which include but are not limited to: anxiety, depress, substance abuse, bipolar disorder or schizophrenia; it's important to detect these disorders and intervene as early as possible. There are two focuses: apps for the consumer and tools for the psychiatrists to support diagnostics. The app that consumers use typically consists of a chatbot that is enhanced with machine learning that analyzes the language and conversation created. The app can respond with scientific evidence based on responses and provide appropriate recommendations for help.

Another case study included in the reading regarding heart failure, creating a base line in machine learning to predict when someone may need medical attention due to a failing heart. Previous telemedicine had predetermined thresholds that would often provide a false alert to users. This would lead to increased medical visits and costs. Machine learning analyzed all aspects of someone with heart failure including, age, smoker or non-smoker status, pacemaker or not, specific bloodwork values, blood pressure, weight, and previously diagnosed chronic illness to determine a true medical visit. Machine learning doesn't replace a human, but it can significantly decrease time wasted and costs.

These are just 3 ways machine learning has integrated its way into life science and biotechnology. Machine learning helps to provide the ill with quicker detection and remedies for their ailments.